SSH Manual: Persistent TOPAS Workflows

High-Performance Computing Protocol - Samuel Barachel Takwirira

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1 Phase I: Persistent Deployment

1.1 1. The Background Launch

REMOTE: SSH Server Session

```
nohup /net/home/samt/topas/bin/topas 2026-01-23_ND3.txt > simulation.log 2>&1 &
```

1.2 2. Detaching the Session

REMOTE: SSH Server Session

```
disown -h %1
```

2 Phase II: Surveillance & PID Management

2.1 1. Locating the Process ID (PID)

If the terminal session was closed, use the global process list to find your ID:

REMOTE: SSH Server Session

```
ps -ef | grep topas
```

2.2 2. Decoding the Process String

In your output (`1233156 ... 01:38:57 topas`), the values represent:

- **1233156**: The unique **PID** (Process ID).
- **01:38:57**: Total CPU time consumed (shows the simulation is active).

2.3 3. Targeted Commands using PID

Once you have identified the PID (e.g., **1233156**), use these specific tools:

- **Search/Verify Status:** `ps -p 1233156`
- **Monitor Live Performance:** `top -p 1233156`
- **Check Open Files/Outputs:** `lsof -p 1233156`

3 Phase III: Data Recovery

LOCAL: M3 Mac Terminal

```
scp user@server:/path/to/results/*.csv ~/Desktop/Results/
```

Emergency Protocols

- **Kill a Specific Process:**

  REMOTE: SSH Server Session

  ```
  kill 1233156    # Orderly shutdown
  kill -9 1233156    # Immediate forced termination
  ```

- **Confirmation of Exit:** If `ps -p [PID]` returns an error or no data, the simulation is no longer running.

4 Re-Acquiring Console Output

If a disowned process needs to be monitored or brought back to the foreground:

4.1 1. Virtual Console (Viewing only)

To view the "live" console output being written to the log:

REMOTE: SSH Server Session

```
tail -f simulation.log
```

4.2 2. Process Hijacking (Re-attaching)

To physically move the process from the background back into your active terminal window, use `reptyr`:

REMOTE: SSH Server Session

```
reptyr [PID]    # e.g., reptyr 1233156
```

4.3 3. Direct Pipe Inspection

If no log file was created, you can tap directly into the process file descriptor:

REMOTE: SSH Server Session

```
tail -f /proc/[PID]/fd/1
```

5 Standard Execution: Background Persistence

To execute TOPAS while ensuring the simulation survives a terminal logout:

5.1 1. Deployment Command

Run the binary using the absolute network path and redirect all streams to a log.

REMOTE: SSH Server Session

```
nohup /net/home/samt/topas/bin/topas 20260127ND3.txt > simulation.log 2>&1 &
```

5.2 2. Session Detachment

Immediately notify the shell to ignore the job upon session termination:

REMOTE: SSH Server Session

```
disown -h %1
```

5.3 3. Surveillance without Interruption

To view the current progress (histories, time steps) as if you were in the original shell:

REMOTE: SSH Server Session

```
tail -f simulation.log
```

*Note: Closing this view with **Ctrl+C** does not affect the background process.*

6 Phase IV: Proper Session Termination

Once the process is disowned and redirected, the terminal session is no longer required.

6.1 1. Formal Exit

Instead of closing the window manually, type the exit command to ensure the SSH buffer clears correctly.

REMOTE: SSH Server Session

```
exit
```

6.2 2. Post-Logout Verification

To confirm the simulation survived the logout, reconnect via SSH and query the system:

REMOTE: SSH Server Session

```
pgrep -a topas    # Confirm PID and Command Path
tail -f simulation.log    # Confirm ongoing calculations. You can of course rename the log file
```